

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Fall 2024
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	05-Sep-2024	Section	BCS- 5D
	Exam Type:	Quiz 1 - Chapter 1	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 1]

1. Which of the following delays is primarily influenced by the distance between the source and the destination?
 - a) Transmission delay
 - b) Processing delay
 - c) Propagation delay
 - d) Queuing delay
2. High traffic intensity in a network often leads to:
 - a) Reduced queuing delays
 - b) Increased queuing delays
 - c) Increased bandwidth
 - d) Reduced propagation delay
3. In a peer-to-peer (P2P) network, the peers are considered part of the:
 - a) Network core
 - b) Network edge
 - c) Both network core and network edge
 - d) None of above

True/False:

- i. Packet switching requires the establishment of a dedicated path between the sender and receiver before communication begins. [T / F]
- ii. Hosts are a part of the network edge, whereas servers are not. [T / F]

Q2. A 4,000-byte packet is sent from a host to a destination through two routers (Router A and Router B). The network path consists of the following links: **[10 marks] [CLO 1]**

- Link 1 (Host to Router A): Bandwidth = 10 Mbps, Propagation delay = 4 ms
- Link 2 (Router A to Router B): Bandwidth = X Mbps, Propagation delay = 6 ms
- Link 3 (Router B to Destination): Bandwidth = 5 Mbps, Propagation delay = 10 ms

Assumptions:

The packet is fully received at each router before being forwarded and there are no additional delays (e.g., queuing, processing).

The total end-to-end delay for the packet is 50 ms, including both transmission and propagation delays across all links.

- a) Calculate the transmission delay on Link 1. (2 Marks)
- b) Calculate the transmission delay on Link 3. (2 Marks)
- c) Determine the required bandwidth X on Link 2 such that the total end-to-end delay equals 50 ms. (6 Marks)